

Cambridge International AS Level
Information Technology 9626 · Paper 1 Theory

Topic 3

Monitoring and control

35 questions from 25 question papers, Feb/March 2022 to May/June 2026.

Questions come first, in date order, then their mark schemes in the same order. Each question is labelled with the paper it came from. Click "Mark scheme" next to a question to jump to its answers, or use the bookmarks panel.

A question that covers more than one topic appears in each of those topics. Topics follow the 2025-2027 syllabus.

Questions

9626/12 Feb/March 2022 · Question 4

[Mark scheme](#)

- 4 Computerised weather stations are used to monitor a number of analogue atmospheric variables.

Describe how **three** of these variables are collected and what needs to happen to the data before it can be used by a computer.

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..... [4]

6 Many houses in cooler countries have central heating systems. In a central heating system, microprocessors are used to control the pump which sends water from the boiler to the individual heaters.

Complete the pseudocode algorithm to show the processing which takes place in this microprocessor-controlled central heating system. The algorithm must prevent attempting to switch the pump on when it is already on and attempting to switch the pump off when it is already off. You may assume `temperature` is the variable representing the actual temperature of the house and `preset` represents the required temperature of the house.

```
REPEAT
INPUT temperature
    IF temperature < preset
        THEN
```

[6]

List examples of sensors the group will use and describe how they will be used with the computer to monitor the pollution level.

[6]

- 6** Sensors are used in monitoring systems to measure physical variables. It is important that sensors are calibrated before use.

- (a)** Describe, in detail, these sensors and give an appropriate use for each. Each use must be different.

Humidity

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Sound

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[4]

- (b)** Explain the importance of calibration.

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[2]

Page 6

8 A microprocessor-controlled car park barrier is used to allow cars to enter a car park. When a car approaches the barrier, the barrier automatically rises to allow the car to enter. It does not lower until the car is safely clear of the barrier system. There is a light sensor and light beam source immediately before the barrier. No tickets are issued by the system. The system operates continuously while it is switched on.

Complete this pseudocode algorithm to describe the processing involved. Your pseudocode algorithm should include IF...THEN statements.

```
REPEAT
    INPUT reading from induction loop
```

..... [7]

- 9 Jose has a greenhouse to grow his plants. He lives in a country which has a warm climate so the greenhouse does not need a heater but has microprocessor controlled windows and a temperature sensor.

(a) Describe how the sensor represents the temperature so the computer can read the data.

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..... [1]

(b) Complete this pseudocode algorithm to show the processing which takes place to control the temperature of the greenhouse.

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WHILE system switched on
  INPUT temperature
  IF temperature > preset
    THEN
      IF window closed
        THEN
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- (c)** Discuss the advantages and disadvantages, to Jose, of using microprocessor control in this scenario.

[8]

[8]

[5]

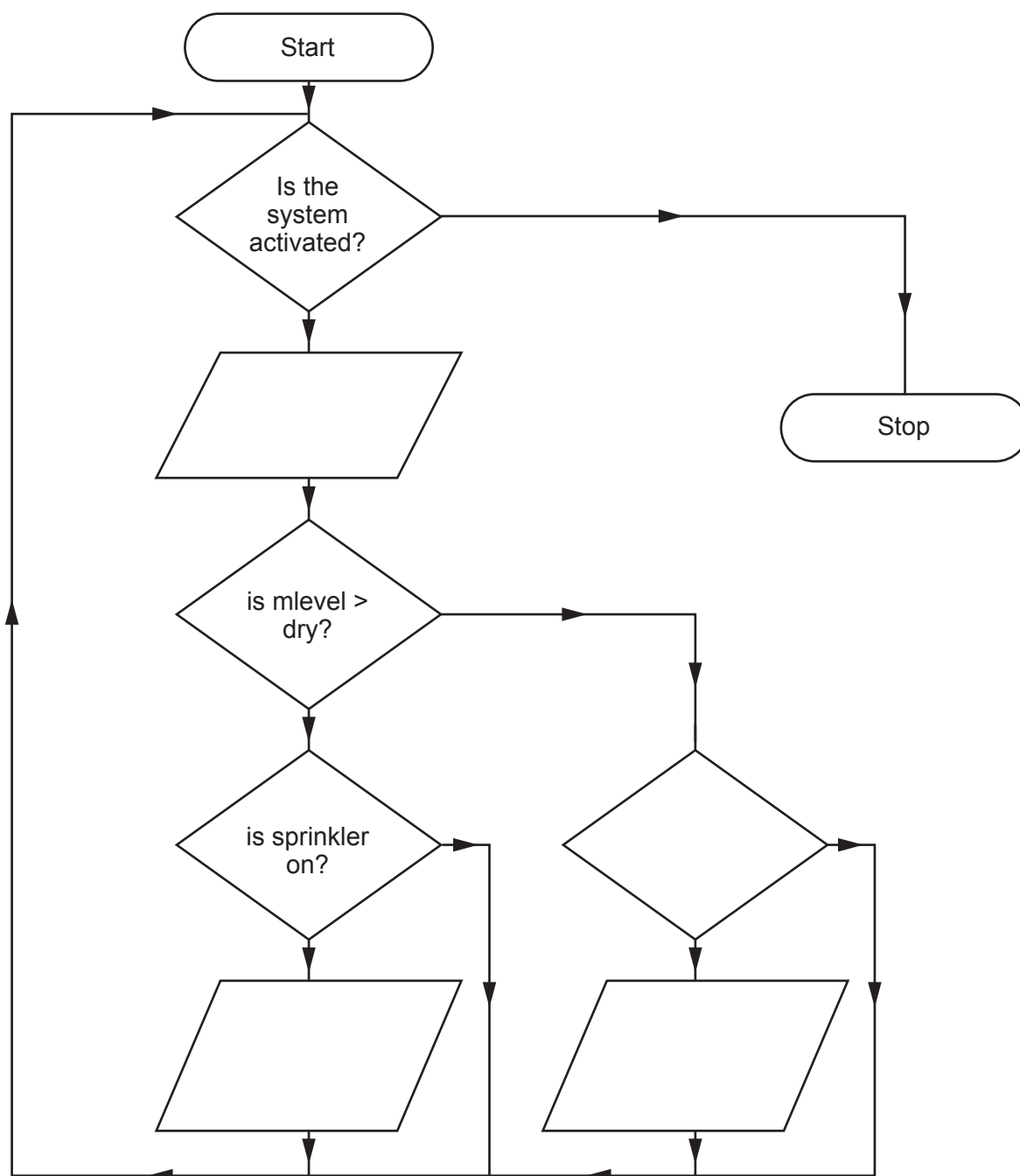
[6]

- 4 Mark has a small glasshouse. He lives in a warm country where plants can die if they are left without water for too long. The glasshouse has a microprocessor controlled sprinkler system and a moisture sensor in the soil.

Complete this flowchart to show the processing required to control the level of moisture in the soil of the glasshouse. You may assume that a pre-set value, **dry**, has already been input so that the system can test if the amount of moisture in the soil has reached or fallen below a certain level.

The user activates the microprocessor system by pressing a switch.

- (a) Complete the flowchart which inputs the moisture reading in the glasshouse and switches the sprinkler on and off according to this value. It also checks if the sprinkler is already on or off before attempting to switch it on or off.



[6]

(b) Flowcharts often contain subroutines.

(i) Draw the symbol used to represent a subroutine in a flowchart.

[1]

(ii) State what you would expect to see in the first box of a subroutine.

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..... [1]

6 Sensors are used to measure physical variables.

(a) Describe these sensors and give an appropriate use in a monitoring system for each.

(i) Light
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..... [2]

(ii) Temperature
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..... [2]

(b) Sensors need to be calibrated before use.

Evaluate the different forms of calibration.

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..... [6]

2 Describe what is meant by a smart home.

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7 A computer system is used to monitor the level of pollution in the air.

Describe how the system would collect, process and display the data.

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6 Sensors are used in monitoring and control to measure physical variables.

- (a)** Describe these sensors. For each sensor, explain how it could be used in a control system. Each system must be different.

Infrared

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Ultrasonic.....

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[4]

- (b)** Control technologies are replacing human-controlled car park barriers and timer-controlled traffic lights.

Discuss the advantages and disadvantages of the use of control technologies in these scenarios.

..... [8]

1 Describe what is involved in data logging.

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[8]

6 Most control technology systems use actuators and sensors.

(a) Describe the purpose of an actuator.

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(b) Describe how an electromagnetic field sensor works when used in a car park control technology system.

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(c) Describe how an ultrasonic sensor works when a driver is parking a car.

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9626/12 Oct/Nov 2023 · Question 4

[Mark scheme](#)

- 4 (a) Describe how touch sensors are used as part of a control system in nuclear power plants.

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- (b) Describe how infrared sensors are used in burglar alarm systems.

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9626/13 Oct/Nov 2023 · Question 5

[Mark scheme](#)

- 5 It is essential to calibrate sensors used in any monitoring technology.

Describe what is meant by:

- (a) two point calibration

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..... [2]

- (b) multipoint calibration.

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[6]

- (a)** Describe how data logging could be used to help the factory owner monitor the environmental conditions around the factory.

[6]

- (b)** Discuss the advantages and disadvantages of monitoring the environmental conditions around the factory using a computer system rather than using manual methods.

[8]

[6]

8 (a) Describe the purpose of **three** sensors that are needed in a microprocessor-controlled burglar alarm system.

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(b) Explain how a microprocessor-controlled burglar alarm system works.

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[8]

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[4]

Mark scheme

[6]

[6]

(a) Describe the use of different sensors to collect data about the weather.

[6]

(b) Describe how the data from the sensors can be processed.

[6]

9626/12 May/June 2025 · Question 5

Mark scheme

5 Describe **five** advantages of using computers rather than a human to monitor water pollution.

..... [5]

9626/12 May/June 2025 · Question 8

[Mark scheme](#)

8 Describe what is meant by an actuator. You must include an example of an actuator in your answer.

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9626/13 May/June 2025 · Question 9

[Mark scheme](#)

9 Describe how a microprocessor-controlled car park barrier system makes use of an induction loop.

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[2]

(b) Compare one-point calibration with multipoint calibration.

[5]

5 Pressure sensors and humidity sensors may be used in a control system.

(a) For each sensor, describe how it gathers data.

(i) pressure sensor

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..... [1]

(ii) humidity sensor

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..... [1]

A car park management system has a pressure sensor in each individual parking space. These pressure sensors are used to find out how many free spaces are available.

(b) Describe how this system works.

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(c) (i) Define the term calibration.

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(ii) Explain why it is important to calibrate pressure sensors.

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9626/11 May/June 2026 · Question 8

[Mark scheme](#)

- 8 A combined heating and air conditioning system has been installed in a house. The system measures the temperature in the house and reacts as programmed. The user can override the temperature settings of the system to suit their needs.

(a) Identify **two** input devices that will be used as part of this system.

input device 1

input device 2 [2]

(b) Explain, with the example of a pump, the purpose of an actuator in this system.

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..... [2]

The system has the following rules:

- If the temperature exceeds 28 degrees Celsius, a fan will turn on and the heating system will turn off.
 - If the temperature falls below 19 degrees Celsius, the fan will turn off and the heating will turn on.
- (c) Complete the pseudocode algorithm to show the processing that takes place in this combined system. The variable `temperature` represents the temperature of the house.

```
REPEAT  
    INPUT temperature
```

Mark schemes

Mark scheme · 9626/12 Feb/March 2022 · Question 4

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Question	Answer	Marks
4	Three from Temperature sensors measure ambient temperature Pressure sensors measure atmospheric pressure Humidity sensors measure absolute and relative humidity Light sensors measure sunlight/light levels Reed switch and tipping bucket measure rainfall Analogue data needs to be converted into digital data/data that a computer understands – 1 mark	4

Mark scheme · 9626/11 May/June 2022 · Question 6

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Question	Answer	Marks
6	Six from: <pre> REPEAT INPUT temperature IF temperature < preset THEN IF pump switched off THEN send signal to switch pump on ENDIF ELSE IF pump switched on THEN send signal to switch pump off ENDIF ENDIF UNTIL system switched off </pre> 1 mark for correct 1st IF statement 1 mark for correctly positioned first action 1 mark for correctly positioned 1st ENDIF 1 mark for correct 2nd IF statement 1 mark for correctly positioned second action 1 mark for correctly positioned 2nd ENDIF 1 mark for correctly positioned 3rd ENDIF 1 mark for correct UNTIL statement... ...1 for correct position of correct UNTIL statement	6

Question	Answer	Marks
10	Six from: Temperature, light, turbidity, pH, O₂, CO₂, nitrite, nitrate (must have two) Place sensors in the river next to the farm The sensors would be connected to ADC/computer Sensors send data to analogue to digital convertor ...enables the computer to understand the data Computer receives data from ADC Readings from sensors are compared with normal river values <u>by the computer</u> Differences/results are printed out/displayed on screen Graphs are <u>automatically</u> produced by computer showing values from river and normal values Computer stores readings in a table/database/file/spreadsheet ready for further processing	6

Question	Answer	Marks
6(a)	Humidity sensor combination of moisture sensor and temperature sensor/measures how much moisture is in atmosphere together with temperature – 1 mark Used in weather stations to measure relative/absolute humidity /air humidity /any suitable example – 1 mark Sound sensor converts sound waves into voltages/electrical signals – 1 mark Used in environment monitoring systems to measure noise pollution /any suitable example – 1 mark	4
6(b)	Two from: The accuracy of all sensors reduces after a period of time... ...due to constant use in/exposure to the atmosphere/liquids that are being used Regular calibration helps to maintain the accuracy of the sensors/helps to make sure sensors produce accurate/correct/reliable readings Other devices within a monitoring system may deteriorate over time resulting in a need for recalibration	2
6(c)	Five from: Only one measurement point/reading with one point calibration whereas with two point calibration at least two readings are taken One point calibration is appropriate for sensors used to measure a value that is constant/never changes... ...whereas two-point calibration is used with sensors that measure a constantly changing variable With one point calibration the offset only is calculated/ the sensor reading is subtracted from the known value only... ...the offset is then added to every subsequent reading With two point calibration sensitivity needs to be included not just an offset Two point calibration is used where there is a linear relationship between the two readings The slope of the line connecting the two readings is compared with the slope of the line of the pre-calibrated sensor/the range values of the sensor is compared with the range values for the standard Two point calibration compensates for both offset errors and sensitivity/slope errors	5

Question	Answer	Marks
8	REPEAT <i>INPUT reading from induction loop</i> IF reading = pre-set value THEN send signal to raise barrier REPEAT IF light sensor activated THEN send signal to lower barrier ENDIF UNTIL light sensor activated ENDIF UNTIL system switched off Seven from: Appropriate IF statement (comparison of reading) Send signal to raise barrier after first IF Appropriate other IF statement (checking if light sensor activated) Send signal to lower barrier after 2nd IF Appropriate position and matching 1st ENDIF Appropriate position and matching 2nd ENDIF Appropriate UNTIL system switched off statement Appropriate loop checking that light sensor has been activated	7

Question	Answer	Marks
9(a)	One from: - Electrical resistance is changed/voltage is generated (depending on the temperature) (1) Electrical signals are generated (which are converted into values (to represent temperature)) (1).	1
9(b)	<pre> WHILE system switched on INPUT temperature IF temperature > preset THEN IF window closed THEN send signal to motor to open window ENDIF ELSE IF window open THEN send signal to motor to close window ENDIF ENDIF ENDIF ENDWHILE </pre> - send signal to open//open window (NOT 'output') (1) - send signal to open//open window as 1st statement (1) - ENDIF closes the IF statement (1) - ALL three ENDIFs in same vertical plane as relevant IFs (within reason) (1) - ELSE (1) - ELSE in same vertical plane as THEN (1) 'IF window open' (or equivalent) ... (1) ... IF window open condition closed with ENDIF (1) - Close window instruction (1) - Send signal to motor to close window appropriately positioned (1) - IF temperature > preset value condition closed with ENDIF (1) - Whole of IF Window open section indented correctly (allow follow through on close window instruction position) (1) - ENDWHILE as final statement (1)	8

Question	Answer	Marks
9(c)	Eight from: Advantages max 6 • Reduced chance of plants dying/wilting etc (1) • Jose could forget to (e.g.) open window//be so busy that he is unable to (e.g.) open window ... (1) ... computers can be permanently switched on and take action regularly/will not forget (1) • Jose can do other tasks (1) • Computers can take action immediately if the temperature gets too hot ... (1) ... there would be a delay before Jose can take action (1) • Readings taken by computers tend to be more accurate ... (1) ... and action taken is therefore more likely to be the correct one (1) (ACCEPT CONVERSE ARGUMENTS ABOUT JOSE'S INACCURACY) • If Jose was physically disabled, he may find it difficult to open window ... (1) ... whereas computer can do it automatically (1) Disadvantages max 6 • Computers can be expensive to buy ... (1) ... whereas it would not cost Jose any money to look after the plants himself (1) • Computer maintenance costs can be expensive ... (1) ... whereas with Jose that is not a problem (1) • It is quite difficult to program computers to take correct action ... (1) ... but Jose would know instinctively//will have expertise (1) • Sensors can deteriorate//become inaccurate ... (1) ... whereas Jose's actions will tend to be more consistent (1) ... and will need recalibration (1) • A computer-controlled system will not be able to function if there is a problem with the computer/there is a power cut/load shedding (1) Max. six marks if bullets/list of points Must have at least expansions or comparisons to achieve six marks in either section	8

Question	Answer	Marks
10	Advantages Max 6 marks e.g - Computers are able to take readings at more frequent intervals ... (1) ... as students take time writing down readings/values (allow once only)(1) - Taking readings manually takes time//time consuming (not slower) ... (1) ... because students have to write down the readings/values (allow once only) (1) - Computers can read several physical variables at the same time ... (1) ... whereas students would only be able to read one at a time (1) - Computers can take continuous readings ... (1) ... whereas students find it difficult to continually take readings/students may need to take breaks (1) - Students wouldn't be able to take readings during school holidays ... (1) ... but computers can take readings every day//computers work 24/7 (1) (Readings taken by computers) tend to be more accurate ... (1) ... and so are more reliable//students can make mistakes when tired/due to human error (1) - Training is not required ... (1) ... to collect accurate data (1) Disadvantages Max six marks e.g. - Monitoring systems/computers can be expensive to buy ... (1) ... but students do not have to be paid (1) - Monitoring systems/computer maintenance costs are expensive ... (1) ... whereas there is none with students (1) - Sensors can lose accuracy over time ... (1) ... which would lead to inaccurate readings being taken (1) - Sensors may not be calibrated properly ... (1) ... so any measurements will be inaccurate (1) Max. six marks if bullets/list of points Max six marks if no attempt at expansion	8

Mark scheme · 9626/13 Oct/Nov 2022 · Question 2

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Question	Answer	Marks
2	Five marks available: • A real-time system has to respond to/process inputs//inputted data instantaneously (1) • An air conditioning system has to deal with inputs continuously ... (1) ... unless the user switches the system off (1) • This type of processing is usually found in systems that use sensors (1) • The user interface of a real-time system may use specialist input devices to provide data input ... (1) ... such as a keypad/touchscreen (1) • Usually deals with small amounts of data//does not deal with a lot of data (1) • Real time systems are such that the output affects the input (1)	5

Mark scheme · 9626/13 Oct/Nov 2022 · Question 7

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Question	Answer	Marks
7	These answers are based on the use of devices that take measurements. Follow a similar procedure for devices such as light beams, or electronic contacts Six marks available: <i>Identification of inputs and purpose</i> e.g. • A keypad to switch the system on/select zones (1) • Infra-red sensors to detect heat from body (1) • Sound sensors to detect the level of sound (1) • Pressure sensors/pads to detect an increase in weight (1) <i>Processing of signal and decision making</i> e.g. • <u>The microprocessor</u> compares the input reading with the pre-set value/criteria ... (1) ... which would have been programmed in (1) ... to check if reading is above pre-set value (1) <i>Outputs as a result of processing</i> e.g. • Instruction/signal/activation sent to actuator//output device//named suitable device (1) • Relevant output from named suitable device (1) • Signal (NOT 'an output') sent to the police//monitoring service (1). Must have at least one from each to get full marks	6

Question	Answer	Marks
4(a)	<pre> graph TD Start([Start]) --> D1{Is the system activated?} D1 -- no --> Stop([Stop]) D1 -- Yes --> Input[/Input mlevel/] Input --> D2{is mlevel > dry?} D2 -- no --> D3{is sprinkler off? R} D2 -- yes --> D4{is sprinkler on?} D4 -- yes --> S1[/send signal to actuator to switch sprinkler off/] D4 -- no --> D3 D3 -- yes --> S2[/send signal to actuator to switch sprinkler on/] D3 -- no --> Stop S1 --> D2 S2 --> D2 </pre> SIX marks available: • Yes/No in correct position off first decision box (1) • Two other Yes/No used (off a decision) (1) – A further Yes/No used (off a decision) (1) • <u>Input MLevel</u> in correct place (1) • <u>Is sprinkler on/is sprinkler off</u> added in correct place (1) MLevel>dry NO branch: • Is Sprinkler OFF – YES – switch <u>Sprinkler ON</u> in correct place (1) OR • Is Sprinkler ON – NO – switch <u>Sprinkler ON</u> in correct place (1) MLevel >dry YES branch • Is Sprinkler ON -YES- switch <u>Sprinkler OFF</u> in correct place (1) OR • Is Sprinkler OFF – NO – switch <u>Sprinkler OFF</u> in correct place (1) For final four mark points, marks may only be awarded where the logic leading up to the decision matches that given here and is shown in full.	

Question	Answer	Marks			
4(b)(i)	<table border="1"> <tr> <td></td><td></td><td></td></tr> </table>				1
4(b)(ii)	ONE from: - The name of the subroutine (1) - Start (1)	1			

Mark scheme · 9626/12 Feb/March 2023 · Question 6

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Question	Answer	Marks
6(a)(i)	One mark for description, one mark for example: - Measures/detects light <u>intensity</u>//<u>how much light hitting a surface</u>//level of light (1) - Any suitable example of use - e.g. Used in weather stations to measure the amount of sunshine - Test match cricket to see if its bright enough to play (1)	2
6(a)(ii)	One mark for description, one mark for example: - Measures temperature <u>value</u> (1) - Any suitable example of use - Used in weather stations to measure the temperature (1)	2

Question	Answer	Marks
6(b)	SIX from: One point calibration Max 2 • Is the easiest method to carry out (1) • Only one point of data is used (1) • Can only be used when measuring set/constant physical variables... (1) – ...therefore has a limited use (1) Two-point calibration Max 3 • Is more complex than one point calibration (accept less complex answer)... (1) – ...because is needed when a range of values is being monitored (1) • Two points of data are used (1) • Needs to be used where a linear relationship between sensor readings and the variable being monitored exists... (1) – ...therefore, cannot be used when the relationship is a curve (1) Multi point calibration Max 3 • Multi-point calibration is the most complex method... (1) – used for non-linear, multipoint relationships (1) • Multiple points of data are used (1) • Multi-point calibration is the most complex method... (1) – ...greater knowledge of maths is needed (1) Max. five marks if all three methods not covered Max. five marks if bullets/list of points	6

Mark scheme · 9626/11 May/June 2023 · Question 2

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Question	Answer	Marks
2	Four from: • Aspects of the home are controlled from Smartphone/Voice control device/computer system/remotely (1) • Devices/appliances in the home are connected (1) • TWO types of devices (in the home) that can be controlled: e.g. – lighting, – heating – air conditioning (accept “temperature” once only) – television – kettle – plugs (1) • Controlled by issuing (voice) commands/setting up routines (1) • The commands can be given by voice/sound/ remote control/ tablet/ smartphone (1) • Uses Wi-fi/home network (1) • Uses internet to access devices when outside the home (1)	4

Question	Answer	Marks
7	Six from: Collection • Sensor(s) are used to gather data (DO NOT ACCEPT ‘COLLECT’) (1) • Naming of TWO physical values that would be gathered: – temperature – pH – CO₂ – O₂ – Gas (BOD) – <u>UV</u> light – infrared (1) Note: where candidate states (e.g.) temperature sensor, this may be taken as identification of a physical variable. Processing • (Collected) data is sent to ADC (1) • Analogue data is converted into digital (data) (1) • Digital(ised) data is sent to microprocessor/computer/CPU • Computer stores readings in a table/database/file/spreadsheet (1) • Naming of TWO results being calculated. E.g. – Max temperature – Min temperature – Average temperature (1) • ONE mark for any description of how one result has been calculated. (1) Display • Computer plots graphs showing trend of pollution variables ... (1st) – ...over a period of time (1) • Computer plots graphs automatically (1) • Computer outputs the results/graphs on screen/printer (1) • Results may (also) be shown on a website (1)	6

Question	Answer	Marks
6(a)	Infrared • Detects heat (1) • Turns infrared electromagnetic waves into an electrical signal (1) e.g. • Used in a burglar alarm – accept any reasonable use (1) Ultrasonic • Detects sound (1) • Turns sound waves into an electrical signal (1) e.g. • It is used in automated car parking//reversing system (1) • Measure distance (1)	4

Question	Answer	Marks
6(b)	Advantages MAX five from • Increased employment for technicians needed to maintain the systems//maintenance (1) and programmers to program the systems (1) (May also be used as a disadvantage - Award ONCE only) • Data is processed more quickly (1) • Reaction times of control technologies is much faster (1) • Computer-controlled traffic lights tend to create fewer traffic jams (1) • Computer-controlled car park barriers lead to fewer/shorter queues (at car park entrances) (1) • Any indication of reduced day to day costs (may be exemplified – e.g. ‘no need to pay anyone to do the job’) (1) • System can operate 24 hours a day//the whole/all day//continuously ... (1) ... without needing to pay overtime (1) • The readings/inputs taken by computers are more reliable/accurate than human readings ... (1) ... the number of car park spaces are more accurately calculated (1) Disadvantages MAX five from • Has increased unemployment for car park attendants (1) (Converse answer may also be given as an advantage – Award ONCE only) • The number of new jobs created tends to be far fewer than the number of old jobs lost (1) • (Control systems) can be expensive to install/purchase (1) • A computer-controlled system will not be able to function if there is a problem with the computer/power outage ... (1) ... leading to possible chaos at traffic lights/traffic jams (1) ... causing longer queues at car park entrances (1) • Computer maintenance costs can be expensive (1) • Sensors can deteriorate after a period of time //suffer wear and tear (1) ... whereas this would not be a problem with car park attendants (1) Max. six marks if uses a bulleted list.	8

Mark scheme · 9626/13 May/June 2023 · Question 1

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Question	Answer	Marks
1	Four from: • Automatic collection/storing of data (1) • Using a computer and sensors to collect data (1) • This data is analysed/processed (1) • The results are output in the form of graphs/charts/tables (1) • Data is collected over a period of time ... (1) ... continuously/at regular intervals (1) • The analysis/processing usually requires special software (1) • Two of the following examples of physical variables logged: temperature sound light pressure (1)	4

Mark scheme · 9626/13 May/June 2023 · Question 5

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Question	Answer	Marks
5	Eight from: • Computers are able to take readings at more frequent intervals ... (1st) ... as humans take time writing down readings (1) • Computers can read several physical variables at the same time ... (1st) ... whereas humans would only be able to read one at a time (1) • Computers continuously take readings ... (1st) ... whereas humans take breaks (accept any reason why an individual human would not take readings continuously) (1) • Computers take readings as planned (as planned is an example) ... (1st) ... whereas humans may forget (1) • Readings taken by computers tend to be more accurate (1) • Computers can produce accurate graphs automatically/instantaneously ... (1st) ... whereas humans take time to draw them (1) • Tables of results are produced automatically after the readings are received by the computer ... (1st) ... unlike a human who would take a lot of time to write them down neatly (1) • Computers can work in dangerous conditions (1st) plus one mark for an example of a dangerous condition (1) Max 6 marks for just bullet points	8

Question	Answer	Marks
6	Eight from: <i>Inputs (accept device OR data) (Candidate has to be clear they are discussing input)</i> • A button for a pedestrian to press//to indicate they wish to cross (1) • Signal from button (1) • Preset value(s) for delay between request and light change ('how long a light stays on' on its own is TV) (1) <i>Processing</i> • System waits for a preset time (or similar description (not 'when time is up')) (1) • Computer sends a signal to the actuator/control box... (1st) ...to change the traffic lights from green to red (1) • Computer sends signal to actuator to change pedestrian control light to safe (1) • Computer initiates count down (1) • After another pre-set time sends signal to actuator to change pedestrian control light to unsafe (1) • Computer sends a signal to the actuator/control box to change the traffic lights from red to green (1) <i>Outputs – (Candidate has to be clear they are discussing outputs)</i> • Display above pedestrian button indicating don't cross/red man/indicating safe to cross/green man (1) • Beeping sound (1) • Red/amber/green light that drivers can see (1) • Count down (1) Must have at least one from each group to get full marks	8

Question	Answer	Marks
6(a)	Two from: • An actuator provides the output from a control system (1) • Is the mechanical part of a control system (1) • An actuator controls/operates a device (1st) • such as opening a valve/activating a motor/switch (1) • It turns an electrical signal (from a microprocessor) into movement/action/physical (1)	2
6(b)	Two marks from: • Measures the change in a magnetic field(1st) • caused by the presence of a metallic/ferromagnetic object/body of the vehicle (1) OR • Identifies a magnetic field from a car (1st) • caused by the car being metal (1) Generic answers: • Are used (at the entrances to car parks) to control barriers (1) • Are used to detect the number/position of spaces available//whether a (parking) space is occupied (1)	2
6(c)	Two from: • A (device in the) car sends out sound waves (1st) • and the sensor receives the sound waves //after they are reflected back (1) • The device calculates the distance from an object (1st) • by measuring the time between emission and reception of a sound wave/signal (1)	2

Question	Answer	Marks
4(a)	Two from: • Touch sensor is used for measuring fluid levels (1st) • such as the cooling water/coolant level of the reactor (1) • A capacitive touch sensor measures the capacitance between two conductors separated by an insulating plate (1st) • One of the conductors will be the fluid (whose level is being measured) (1) • Detects when the fluid is touching the conductor (1)	2
4(b)	Max Two from • Used to detect (presence/movement of) humans/living beings (1) • Human bodies emit heat/thermal energy (1) • By detecting heat/radiation (from heat) (1) OR Max Two from • Used to detect <u>movement</u> (1) • (Infrared) beams can be beamed across (e.g.) the entrance to the room (1) • The infrared sensor detects the beam/detects if it is broken (1)	2

Mark scheme · 9626/13 Oct/Nov 2023 · Question 5

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Question	Answer	Marks
5(a)	Two from: - Two readings/measurements need to be taken (using the sensor) (1) - Can correct slope/offset errors (1) - Best used where readings are linear (1) - Reading taken at the low end of the range and the high end of the range (1) - Readings taken by a check/reference instrument/uses (previously) known readings (1) - Value for offset is calculated (1st) - by comparing to reference readings/known readings (1)	2
5(b)	Two from: - Uses many/more than two readings/measurements (from the sensor) (1) - Readings taken at a high value, a middle value and a low value (1) - Best used where readings form a curve/non linear (1) - Readings taken by a check/reference instrument/uses (previously) known readings (1) - The calibration cannot be carried out by just adding an offset value/by allowing for the slope (1) - To calibrate the sensor the algorithm needs to include 'curve fitting' (1) - Uses a formula to calculate the correction (1) - Uses a quadratic function such as $y = ax^2 + bx + c$ (1st) - where y is the reading from the standardised sensor/actual value x is the reading from the sensor needing calibration (1)	2

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Question	Answer	Marks
5	Six from: - The required temperature is set by the user - The required temperature is input using a remote control/touchscreen/number pad - A sensor is used to read/gather/monitor the (current) temperature - The data collected by the sensor is analogue (shows clear awareness that original data is analogue) - The data is converted to digital via an ADC Microprocessor compares this data/temperature to the required/input value (this must be stated and not implied) - If temperature is, nothing happens - If the data is different than the required value, the microprocessor sends a <u>signal</u> ... (1st) ...to the actuator (1) If lower, the microprocessor/actuator switches the heater on If higher, then the microprocessor/actuator switches the heater off - Process is constantly repeated	6

Question	Answer	Marks
6(a)	Six from: • Sensor used to gather data (1) • Identifies ONE suitable sensor for collecting information on presence OR absence of pollution (1) • Data collected is analogue (any awareness) (1) • Analogue to Digital Convertor converts data to digital (1st) • So can be used/understood by the computer (1) • Computer analyses/processes data (1) • Results presenting in appropriate format/tables/graphs/charts (1)	6

Question	Answer	Marks
6(b)	Eight from: <i>Advantages MAX SIX:</i> • Data is continually monitored//no reason for monitoring not to occur (may exemplify) (1st) • if an event happens it should not be missed (1) • Computer systems can monitor multiple sensors simultaneously (1) • The logged data is already in electronic form (1st) • no transcription errors/no human error (1) • Does not need to be entered/typed in (1st) • which saves time (1) • Personnel can do other useful tasks (1) • Data is immediately sent to the computer (1st) • The data can be displayed immediately/in real time (1) • The data can be analysed immediately/in real time (1) • Can monitor 24/7 (1) • No risk to humans/equivalent answer (1) • Can be monitored remotely (1) <i>Disadvantages MAX SIX:</i> • Any interruptions to the power supply could cause sensor to stop working//collection of data to be missed/not happen (1) • Any damage/failure of the equipment could cause collection of data to be missed/not happen/be inaccurate (1) • The maintenance of equipment can be expensive/dangerous/time consuming//requires expertise/training. (1) • Data logger can not react to emergencies//unusual circumstances (1)	8

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Question	Answer	Marks
9	Six from: • Sensor identifies approach/presence of vehicle (1) • Identifies ONE suitable sensor (Induction loop/pressure sensor/light sensor) (1) • Microprocessor registers presence of vehicle (1) • Microprocessor checks if barrier is raised/lowered (1) • If barrier down it needs to be raised (1) • Microprocessor sends a signal to an actuator/motor (1st) • which then raises the barrier (1) • If barrier up microprocessor does nothing (1) (if mark not awarded for MP 5, a mark may be awarded for “If barrier up”) • Computer checks to see whether vehicle is present/has gone (1) and if vehicle not detected then the barrier comes down (1)	6

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Question	Answer	Marks
8(a)	THREE marks available e.g. • Motion sensor detects movement (1) • Infrared sensor to detect heat (1) • Sound sensor to detect level of sound/unexpected sound (may exemplify) (1) • Pressure sensor (placed under a rug/carpet) to detect increase in weight by burglar standing on it (1) • Light sensor to detect a break in a light beam. (1)	3
8(b)	THREE from: • Microprocessor may be programmed to have (a range of) acceptable levels (1) • If sensor is activated/sensor readings are outside the expected levels/range (1) • Data is sent from sensor to Analogue to Digital Converter (ADC) (1) • ADC converts the data to digital (1st) – so can be understood by the microprocessor (1) • It sets off an audible alarm sounds/activates lights/alerts police or alarm company or key holder. (1)	3

Question	Answer	Marks
9	EIGHT marks available EIGHT from: Pros max SIX: - Computers never forget to take readings (1) - They take readings constantly (accept 24/7) (1) - No need for human intervention to take readings (1) - The readings are more accurate//can detect readings that humans may not (1) - Readings of several variables can be taken simultaneously (1) - Computers can take readings in dangerous areas for humans to be (1) - Transmission of data can be in real time (not quicker) (1) - Data can be stored for later collection to save electrical power/to reduce human impact (1) Cons max 6: - The monitoring equipment will have to be placed by humans (1) - Sensors/computers have to be repaired/maintained by humans (1) - Environment can affect the signal (1) - Geographical events can affect or move the equipment (1) - The presence/maintaining of the system can affect pollution itself (1) - Humans still have to plan/decide where to place sensors and which sensors to use/which variables to monitor. (1) Conclusion - Valid conclusion based on points made (1)	8

Question	Answer	Marks
3	EIGHT marks available: Identification of control technology (1st mark) (Max FOUR) e.g. • (Smart) traffic lights (1st) • Smart/reactive motorways (1st) • Pedestrian/user controlled lights (1st) • Pollution speed limits (1st) • Lane speed/closure displays/reactions (1st) Evaluation of (2nd mark) (Max SIX) e.g. (Smart) traffic lights • Fewer traffic jams (than preset controlled) (1) • Reduces waiting time (1) • Reduces accidents (1) • Drivers more likely to respect the lights (1) • Cost advantage/disadvantage (1) • Dependant on camera (1st) • To track traffic (1) • That may not be working (1) • Requires maintenance/recalibration (1) • Requires power//will not work if there is a power outage (1) Pedestrian controlled lights e.g. • Traffic flow only interrupted when a pedestrian requests to cross (1) • Smoother traffic flow (1) • Reduces accidents (1) • Fake pushes still stop traffic/no guarantee that anyone actually wants to cross the road (1) Conclusion • Overall, conclusive/evaluative statement about control technologies (1)	8

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Question	Answer	Marks
11	FOUR from: • Calibration improves the accuracy of sensors after they have become inaccurate (1) • Can correct the sensor so original reading is accurate (1) • (OR) can apply an offset to take account of an inaccurate reading (1) • Only one measurement point is needed (1) • Appropriate for sensors that have to measure a single type of value/variable (1) • Does not need to be calibrated against any other range of parameters (1) • Often needed for sensors being used at extreme values (1) • Reading is taken (from the sensor) and compared with a known value (1st) – in the range being measured (1) • Sensor reading is subtracted from the known value (1st) – which gives the offset (1) • The algorithm uses the offset by adding it to every reading in the parameter range being measured. (1)	4

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Question	Answer	Marks
4	One mark per bullet point to a maximum of six marks • They can cause a loss of manual skills (1) • They can stop people from performing simple exercise (1) • They can lead to people becoming lazy (1) • They could become over reliant on microprocessor-controlled devices (1) • The system will not function if there is a problem with the Wi-Fi/internet (1) • The system will not function if there is a power cut (1) • They can be vulnerable to hackers (1) • All devices have to be on the same network (1) • Can increase electricity use (1) • Automatic running of routines may have undesired affect (1)	6

Question	Answer	Marks
7(a)	One mark per bullet point to a maximum of six marks. • A moisture/humidity sensor/tensiometer (1st) – is used to measure absolute and relative humidity (1) – measures the amount of moisture in the air (1) • A wind sensor/anemometer (1st) – is used to measure the direction of the wind (1) – used to measure strength of wind (1) • A raindrop/raingauge/rain sensor (1st) – is used to measure rainfall (1) • A light sensor/photodetector (1st) – is used to measure/sense light (1) • A pressure/barometric sensor (1st) – is used to measure pressure values and height changes. (1) • A temperature/NTC thermistor/RTD thermocouple/IC sensor (1st) – is used to measure temperature (1)	6
7(b)	Six marks from: • The data will be sent to the computer for processing (1) • Via Analogue Digital Convertor (ADC) (1) • ADC will convert the analogue data to digital (1st) - So, the computer can read it (1) • It can create graphs/charts (1st) – To show monthly trends (1) • It can do calculations (1st) – Such as averages/max min (1) • It can produce predictions (1st) – Based on historical of weather data (1) • It can produce reports of weather trends (1st) – Such as global warming (1) • It can produce weather warnings (1st) – From its collected data. (1)	6

Mark scheme · 9626/12 May/June 2025 · Question 5

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Question	Answer	Marks
5	One mark per bullet point to a maximum of five marks. - Computers can collect readings (from sensors) at more frequent intervals (1) - Computers are capable of accessing more than one variable simultaneously (1) - Computers work any time, day or night//24/7 (1) - Computers accept readings at precise/regular intervals (1) - Computers can be programmed to automatically (NOT immediately) take action on readings (1) - Computers will always accept the readings/do not forget (1) - It is safer to have a computer take readings in extreme conditions (1) - No need to input the data manually//avoids data input/recording mistakes (1) - Improved processing speed (1) - Improved accuracy of processing (1)	5

Mark scheme · 9626/12 May/June 2025 · Question 8

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Question	Answer	Marks
8	Max one from: - An actuator converts energy into action//mechanical motion (1) - Performs a movement//physical action (1) - An actuator reacts to a signal to make something move (1) Max one from: EXAMPLE - linear - rotary - soft - hydraulic - pneumatic - electric - thermal - magnetic - mechanic.	2

Mark scheme · 9626/13 May/June 2025 · Question 9

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Question	Answer	Marks
9	One mark per bullet point to a maximum of five marks. • A vehicle passing over the loop causes a change in inductance (1) • The change in inductance is used to detect the vehicle (1) • The metal in the vehicle causes a change in the magnetic field (1) • The change in the magnetic field causes a current to flow (1) • Loop sends back data which is converted to digital (1) • Computer compares data to pre-set value (1) • If computer detects a change, (1st) – it sends signal to the actuator (1) • The actuator will raise/lower the barrier. (1)	5

Mark scheme · 9626/11 and 9626/13 Oct/Nov 2025 · Question 5

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Question	Answer	Marks
5(a)	Two from: • Sensor accuracy changes with time/use because of wear and tear from environmental circumstances/constant use (1) • Sensors might not be adapted to the required environment because the conditions can change (1) • Calibration maintains the accuracy of the sensors to ensure they produce reliable/accurate/correct readings (1) • New sensors need to be setup/calibrated to ensure they meet the appropriate requirements because they will be set to factory settings//have preset parameters (1) • Other components that are part of the system/impact on the reading may affect the reading so making it inaccurate (1)	2
5(b)	Five from: e.g. • Only one measurement point/reading with one-point calibration//with multipoint calibration at least three readings are taken (1) • One-point calibration is used with sensors that measure a single value// multipoint calibration is used with sensors that measure a range of values that have a non-linear relationship (1) • One-point calibration requires a simple calculation to take account of the error//multipoint calibration requires a complex calculation (1) • They both mitigate/reduce impact of structural errors in readings//inaccurate readings (1) • They both need to be done regularly (1) • They both compare values with known/correct/accurate measurements (1)	5

Question	Answer	Marks
5(a)(i)	ONE mark available: - The force applied is measured (1) - Uses the piezoelectric effect (1) - The force applied to its surface creates an electrical signal (1) - Electronic resistance changes when a weight is applied (1)	1
5(a)(ii)	ONE mark available: - Affected by BOTH temperature and moisture (1) - Gathers changes in electrical impedance/capacitance/temperature (1) - A metal oxide strip resistance changes with humidity (1) - Impedance changes with ions in salts (1)	1

Question	Answer	Marks
5(b)	EIGHT marks available: - The standard pressure (without a car) is measured/has a preset value (1) - Data is captured constantly (do not award sensors <u>send</u> data for this mark point) (1) - Converted to digital (1st) by an ADC (1) - The computer/processor compares the readings to the standard pressure reading//a preset value (1) - If there is a pressure increase (from the standard), a car is present (in the space) (1st) This is recorded/reflected in the database (1) - If there is no change from the standard pressure reading, the space is empty (1st) This is recorded/reflected in the database (1) - Number of free spaces is calculated (1) - The number of free spaces/vehicles present is displayed (DO NOT ACCEPT alternative outputs that are not to do with finding out how many free spaces are available) (1)	8
5(c)(i)	ONE mark available: An adjustment to a sensor (reading)//tuning a sensor to make it accurate (1)	1
5(c)(ii)	TWO marks available: - Sensors become inaccurate (1) - Calibration takes account of inaccuracy (1) - Increases the active/working life of the sensors (as long as they are sending a reading) (1) - So that readings are (always/remain) accurate (1)	2

Question	Answer	Marks	Guidance
8(a)	TWO marks available: • Temperature sensor/Thermostat (1) • Keypad (1) • Microphone (1) • Touch screen (1) • (Push) Switch (1) • Remote control (1)	2	Do Not Accept: • Smart Phone etc. • Button • Heat sensor
8(b)	TWO marks available: • The actuator converts a signal <u>from the microprocessor</u> into an action/to perform an action (1) • To turn the pump on/off (1)	2	

Question	Answer	Marks	Guidance																						
8(c)	EIGHT marks available: <table><tr><td>IF temperature >28</td><td>1</td></tr><tr><td>THEN (on separate line from the IF) (award if EITHER THEN fits this mark point)</td><td>1</td></tr><tr><td>(Send signal to) turn on fan ONLY AWARD IF THIS IS AS A CONSEQUENCE OF TEMPERATURE >28 <u>ONLY</u></td><td>1</td></tr><tr><td>Send signal to turn on indented</td><td>1</td></tr><tr><td>ENDIF correctly positioned within the algorithm</td><td>1</td></tr><tr><td>IF temperature <19 with matching ENDIF</td><td>1</td></tr><tr><td>THEN (on separate line from the IF) (Send signal to) turn heating on ONLY AWARD IF THIS IS AS A CONSEQUENCE OF TEMPERATURE <19 <u>ONLY</u></td><td>1</td></tr><tr><td>Any use of ENDIF when in capitals and not split</td><td>1</td></tr><tr><td>UNTIL SYSTEM SWITCHED OFF</td><td>1</td></tr><tr><td>UNTIL statement correctly positioned and indented</td><td>1</td></tr><tr><td>ALL IFs and ENDIFs correctly indented</td><td>1</td></tr></table>	IF temperature >28	1	THEN (on separate line from the IF) (award if EITHER THEN fits this mark point)	1	(Send signal to) turn on fan ONLY AWARD IF THIS IS AS A CONSEQUENCE OF TEMPERATURE >28 <u>ONLY</u>	1	Send signal to turn on indented	1	ENDIF correctly positioned within the algorithm	1	IF temperature <19 with matching ENDIF	1	THEN (on separate line from the IF) (Send signal to) turn heating on ONLY AWARD IF THIS IS AS A CONSEQUENCE OF TEMPERATURE <19 <u>ONLY</u>	1	Any use of ENDIF when in capitals and not split	1	UNTIL SYSTEM SWITCHED OFF	1	UNTIL statement correctly positioned and indented	1	ALL IFs and ENDIFs correctly indented	1	8	
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